Report 1: Project Description Report

Hospital Management System (HMS)

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1. Business Idea & System Purpose

The proposed **Hospital Management System (HMS)** is an enterprise software solution designed to digitize, centralize, and streamline core administrative and clinical workflows within modern healthcare institutions. By automating patient record handling, consultation appointments, diagnosis tracking, and financial billing, the system enhances patient care quality and eliminates operational inefficiencies caused by traditional paper records.

Problem Statement

Healthcare facilities operating on paper-based workflows face severe operational challenges: long patient waiting times, misplacement of medical histories, double-booked consultation schedules, and frequent accounting errors in patient billing.

System Purpose

To establish a centralized relational data environment that guarantees instantaneous access to patient records, optimizes doctor consultation schedules, automates billing calculations, and provides complete operational transparency.

2. Project Scope

In-Scope Functions (Covered)

- Patient registration and demographic profiling
- Doctor profile management & schedule availability
- Appointment booking, rescheduling, and cancellation
- Medical history, diagnosis, and prescription recording
- Invoice computation and payment processing

Out-of-Scope Functions (Excluded)

- Pharmacy inventory & vendor supply management
- Direct integration with ICU or lab hardware devices
- External health insurance provider API connectivity
- Hospital staff HR management and payroll processing

3. Core System Classes & Requirements

Class Name	Primary Responsibility / Domain Scope
Patient	Stores personal identity, contact attributes, demographic information, and provides access to historic medical profiles.
Doctor	Maintains physician credentials, medical specializations, consultation working hours, and diagnosis authorization.
Appointment	Handles scheduling requests, manages time slot reservations between doctors and patients, and tracks booking statuses.
MedicalRecord	Logs doctor diagnoses, clinical observations, treatment plans, and electronic prescriptions per visit.
Billing	Calculates consultation fees, generates official invoices, processes payments, and updates transaction statuses.